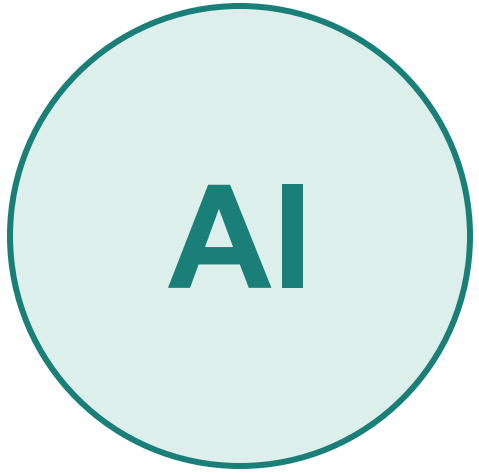
INTRODUCTION TO

Artificial Intelligence & Machine Learning

A visual, practical overview



Concepts • Learning types • Limits

INTRODUCING Dr. Elfikky

Google Scholar SIGN IN



Abdelrahman Elfikky

FOLLOW

University of Arkansas at Little Rock, USA.
Verified email at ucsc.edu - Homepage

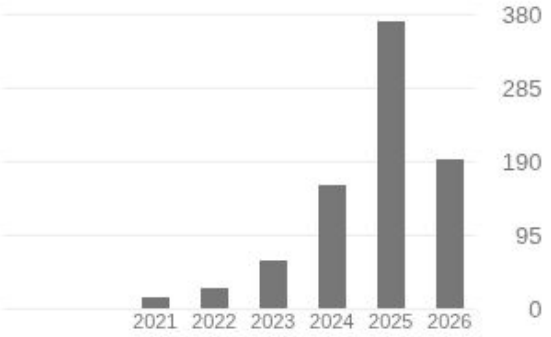
Optical Communications Quantum Autoencoder Machine learning Large Language Models Agentic AI

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TITLE	CITED BY	YEAR
Coding techniques for diversity enhancement of dense wavelength division multiplexing MIMO-FSO fault protection protocols systems over atmospheric turbulence channels EE Elsayed, MR Hayal, I Nurhidayat, MA Shah, A Elfikky, AI Boghdady, ... IET optoelectronics 18 (1-2), 11-31	136	2024
Modeling and investigation on the performance enhancement of hovering UAV-based FSO relay optical wireless communication systems under pointing errors and atmospheric … MR Hayal, EE Elsayed, D Kakati, M Singh, A Elfikky, AI Boghdady, ... Optical and Quantum Electronics 55 (7), 625	107	2023
A survey of deep learning based NOMA: State of the art, key aspects, open challenges and future trends SAH Mohsan, Y Li, AV Shvetsov, J Varela-Aldás, SM Mostafa, A Elfikky Sensors 23 (6), 2946	90	2023
Underwater visible light communication: recent advancements and channel modeling: A. Elfikky et al. A Elfikky, AI Boghdady, S Mumtaz, EE Elsayed, M Singh, ... Optical and Quantum Electronics 56 (10), 1617	62	2024
SAC-OCDMA-FSO communication system under different weather conditions: performance	52	2021

Cited by

	All	Since 2021
Citations	836	832
h-index	15	15
i10-index	18	18

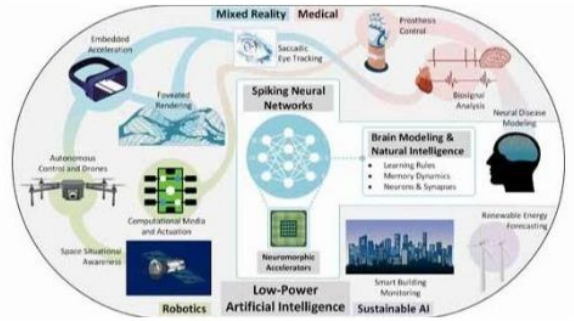


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Received the Outstanding TA Award for ECE Department 2022/2023 year among more than 50 graduate students.



snnTorch reaches 100K+ downloads

Prof. Eshraghian's Python library creating brain-inspired spiking neural networks and a machine learning method for efficient data processing has been widely used for research and development.

Learn more



Outstanding Teaching Assistant Award

ECE PhD student Abdelrahman Elfikky was awarded the ECE Outstanding TA Award for the 2022-23 academic year upon recommendation of his course instructors and students. Congratulations on this recognition, Abdelrahman!

Selected by the Teaching and Learning Center as an Electrical and Computer Engineering department representative to participate and receive the award in the 2024 cohort of the Graduate Pedagogy Fellows.

Baskin Engineering at UCSC
7,406 followers
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Huge congratulations to Abdelrahman Elfikky, a #BaskinEngineering postdoctoral researcher in electrical and computer engineering, who received the 2024 University of California, Santa Cruz Excellence in Teaching Award!

Traditionally reserved for faculty and lecturers, Elfikky is the first and only graduate student to receive this prestigious, campus-wide recognition since the award's inception in 1996.

Elfikky was also the recipient of the Outstanding Teaching Assistant Award for 2022–23 within the electrical and computer engineering department. To date, he is the only graduate student at UC Santa Cruz to receive both a departmental and a campus-wide teaching award.

“He genuinely cares for each of his students and his discussion sections were insanely good," said one of Elfikky's students. "He is truly the most knowledgeable, caring, and professional teaching assistant that I have ever had in my four years at UCSC."

Elfikky earned his Ph.D. in December of 2024 with a research focus on machine learning frameworks for space optical communications. He was presented the award during Chancellor Cynthia Larive’s Excellence in Teaching Award Ceremony on June 6, 2025.

Research areas

From foundational models to field-ready systems.

We reduce computational and communication costs while maintaining accuracy, reliability, and real-world usability.

1 Communication-Efficient Federated Learning and LoRA Fine-Tuning

Bandwidth-aware collaborative AI for wireless, edge, and next-generation 6G networks.

2 Compressed Large Language Models and Agentic AI

Capable AI agents that reason, plan, and use tools on resource-constrained platforms.

4 Autoencoders for Wireless Communications and Security Operations

End-to-end learning for wireless systems and practical anomaly detection for modern security operations.

5 AI and Autoencoders for Space Optical Communications

Learning-based transmitters and receivers for reliable satellite and aerospace optical links

Research support

Partnerships that move research into practice.

$264K+

research funding, gifts, equipment, and in-kind support

NVIDIA Academic Grant

$155K in-kind

Blackwell GPUs, deployment kits, and infrastructure for LLM deployment, Physical AI, and advanced AI education.

Xie Operations Industry Gift

$50K

Unrestricted support for Agentic AI, cybersecurity, quantum education, student research, and workforce development.

Fortinet Support

$9.6K

Equipment, training, FortiGuard resources, and developer access for AI-driven network threat detection.

Arkansas Space Grant / NASA EPSCoR

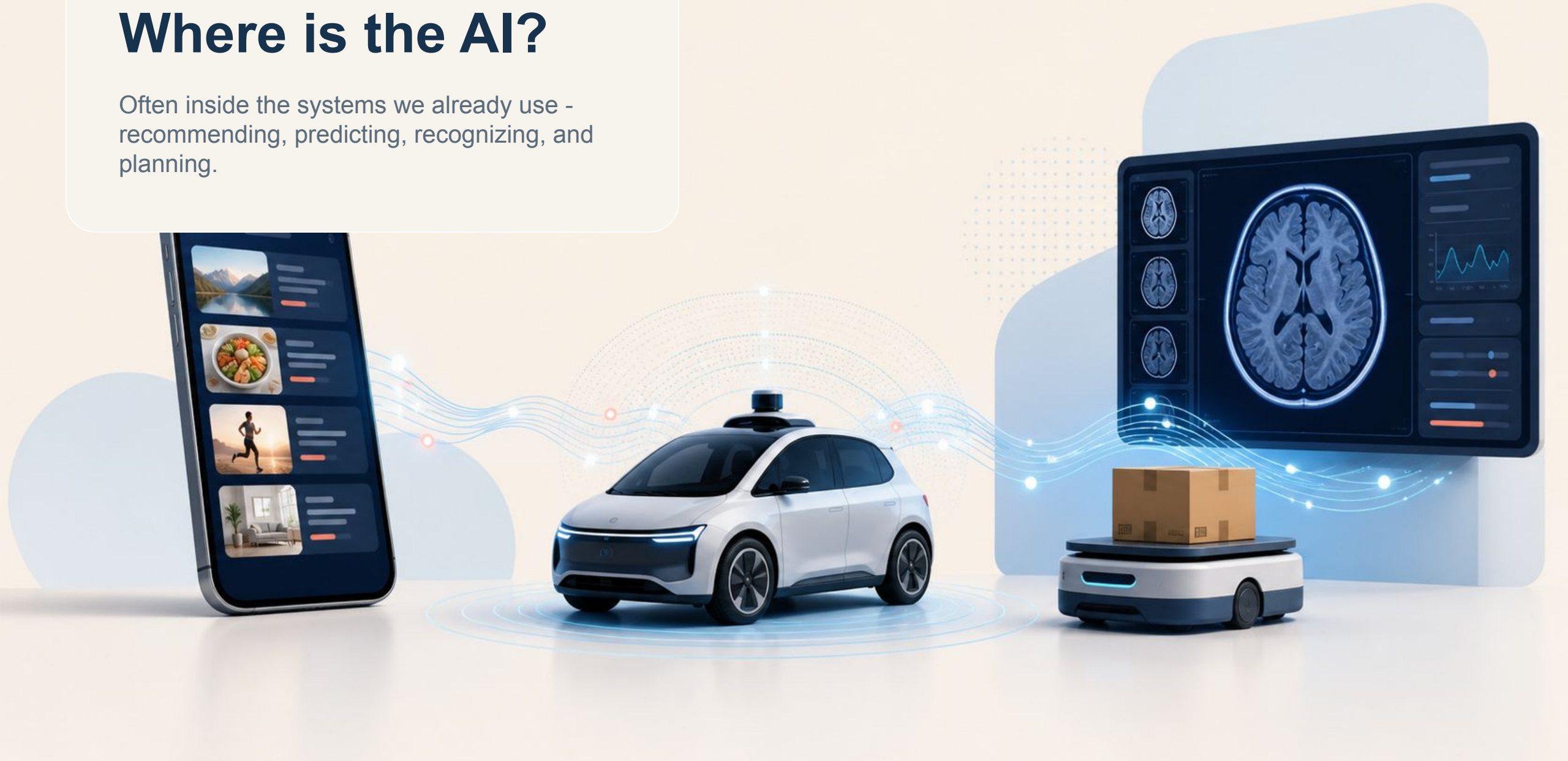
$50K

Learning-based channel estimation and symbol detection for space optical links under atmospheric turbulence.

For more info visit: https://elfikky.com/

Where is the AI?

Often inside the systems we already use - recommending, predicting, recognizing, and planning.



AI is already part of everyday life

The application changes; the underlying jobs repeat.

Recommend

Streaming, shopping, news, and navigation surface likely next choices.

Recognize

Faces, speech, objects, fraud patterns, and anomalies become machine-readable.

Predict

Traffic, demand, risk, and medical signals are estimated from data.

Act

Robots, vehicles, and game agents choose actions toward a goal.

AI is not one product. It is a family of ways to solve problems.

Artificial intelligence extends what machines can do

AI enables computers and machines to perform tasks associated with human intelligence.

Learn

Improve from examples or experience.

Understand

Interpret language, images, sound, or context.

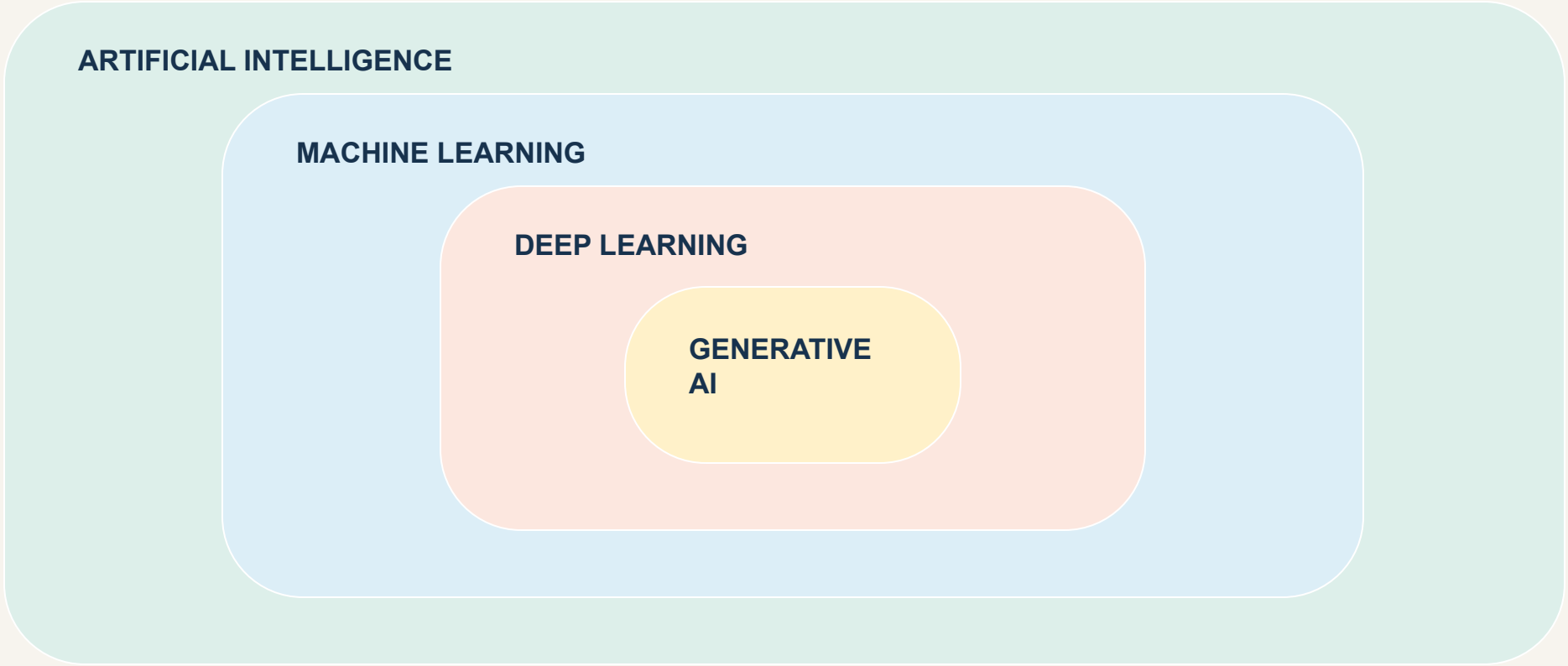
Decide

Choose actions using goals, rules, or predictions.

Create

Generate text, images, audio, code, or plans.

AI is broader than ML, deep learning, and generative AI



Examples: search and logic • regression and trees • neural networks • LLMs

AI algorithms ask different kinds of questions

FAMILY	MAIN QUESTION	SIMPLE EXAMPLE
Search	What sequence of actions reaches the goal?	A delivery robot finds a route
Reasoning / logic	What follows from known facts?	A rule-based diagnosis
Uncertainty	What is the best decision with incomplete information?	Weather-aware planning
Optimization	Which option is best among many?	Scheduling classes or routes
Machine learning	What pattern can be learned from data?	Classifying spam

Every AI problem needs inputs, choices, a goal, and a result

1

INPUT

What the system knows

Map + current location

2

ACTIONS

What it can do

Move, turn, or stop

3

GOAL + COST

What good looks like

Safe, short, on-time route

4

OUTPUT

The selected decision

A route or next action

Example: a delivery robot turns a map into a safe next move.

PART 1

Machine learning begins with a well-defined problem

A clean mental model before the details

Before choosing a method, define the ML problem clearly

1

What is the system given?

Inputs - also called features.

2

What should it produce?

An output: class, number, sequence, or action.

3

Is the correct answer known during training?

If yes, the answer is a target or label.

4

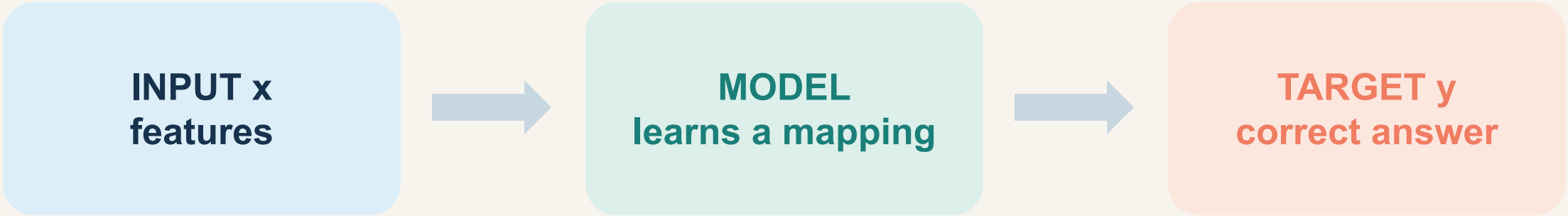
How will success be measured?

Choose an evaluation criterion before training.

Problem definition comes before model selection.

Supervised learning connects inputs to known answers

Training data contains pairs: input x and target y.



Goal: predict y for a new x.

Classification predicts a category

Binary classification

Two possible classes.

Examples: spam / not spam; disease / no disease.

Multi-class classification

More than two classes.

Examples: cat / dog / bird; topic categories.

Training goal

Learn from labeled examples, then assign a class to a new input.

Output y comes from a small, finite set of labels.

Regression predicts a number

The target can take continuous, real-valued values.

HOUSE

price

WEATHER

temperature

DELIVERY

arrival time

ENERGY

demand

Classification asks “which class?” Regression asks “what value?”

Unsupervised learning looks for structure without labels

The system receives data x, but no correct answer y.

Clustering

Group similar samples together.

Example: discover customer segments.

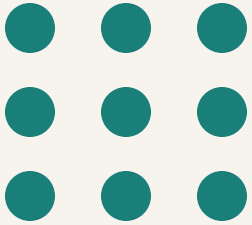
Dimensionality reduction

Represent complex data with fewer dimensions while preserving useful structure.

Pattern discovery

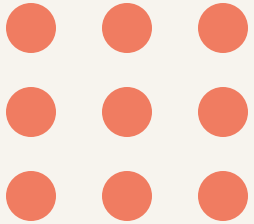
Reveal relationships, density, or organization that were not labeled in advance.

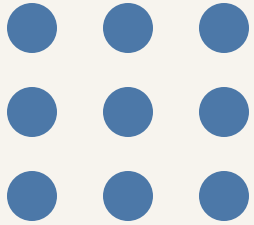
Clustering groups samples according to similarity



Hard clustering

Each sample belongs to one cluster.





Soft clustering

A sample can partly belong to multiple clusters.

Dimensionality reduction makes complex data easier to work with

HIGH-DIMENSIONAL DATA

Many features

REDUCE

Keep useful structure

LOW-DIMENSIONAL VIEW

Visualize or model

Preserve information that matters for the task - not simply the most dimensions.

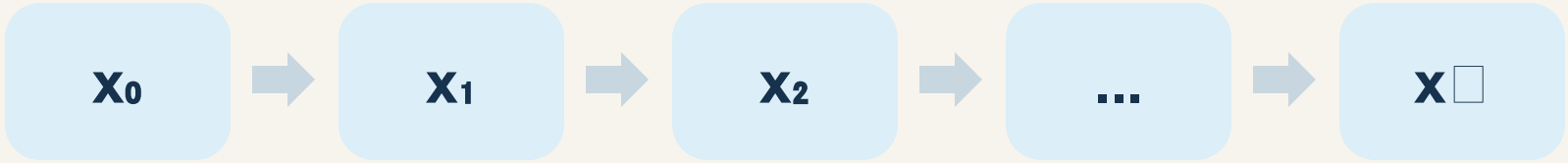
Reinforcement learning improves behavior through reward



No direct signal says which action is best; consequences provide the feedback.

Sequence learning maps ordered inputs to ordered outputs

Order matters: language, audio, time series, and sensor streams unfold step by step.



A hidden state carries context forward; the model learns how to update it and produce outputs.

Examples: translation • speech recognition • next-word prediction

Match each scenario to its learning type

1. **Emails paired with spam labels**
2. Unlabeled customer records grouped by similarity
3. A robot rewarded for reaching a destination
4. Hidden words predicted from surrounding text
5. Candidate schedules improved by selection and mutation

Answer key: the feedback signal reveals the category

Spam-labeled emails	**Supervised learning**	Labels provided
Customer groups	**Unsupervised learning**	No labels
Reward-based robot	**Reinforcement learning**	Rewards + actions
Hidden-word prediction	**Self-supervised learning**	Labels generated from data
Selection + mutation	**Evolutionary search**	Candidate solutions evolve

PART 2

AI is broader than machine learning

A clean mental model before the details

AI can work without training

Search

A* finds routes using costs and heuristics.

The rules are designed; no model must be trained.

Logic + planning

Rules infer facts; planners choose actions that reach a goal.

Optimization

A process searches for the best feasible solution under constraints.

Machine learning

Examples change model parameters; the learned model makes predictions.

Key idea: AI is not the same thing as training. ML learns from data.

Six questions complete an AI problem description

1 Problem class

What data and queries define the task?

2 Assumptions

What is known about the data or solution?

3 Evaluation

How will answers and overall performance be measured?

4 Model type

What aspects will be represented and used?

5 Model class

Which family of candidate models is allowed?

6 Algorithm

What process fits the model or makes decisions?

Natural and artificial intelligence solve problems differently

Natural intelligence

Biological intelligence in humans and animals

Learns continuously from lived experience

Combines perception, memory, reasoning, emotion, and action

Uses common sense and physical-world experience

Artificial intelligence

Human-made systems designed for intelligent tasks

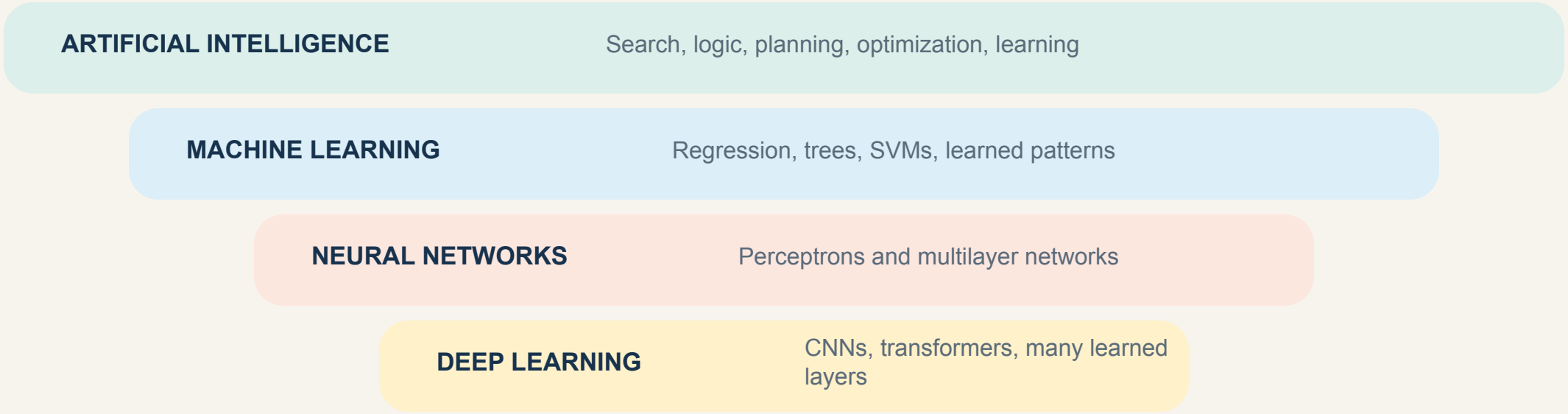
May use rules, search, optimization, or learned models

Processes information at high computational speed

Can exceed humans on specific, well-defined tasks

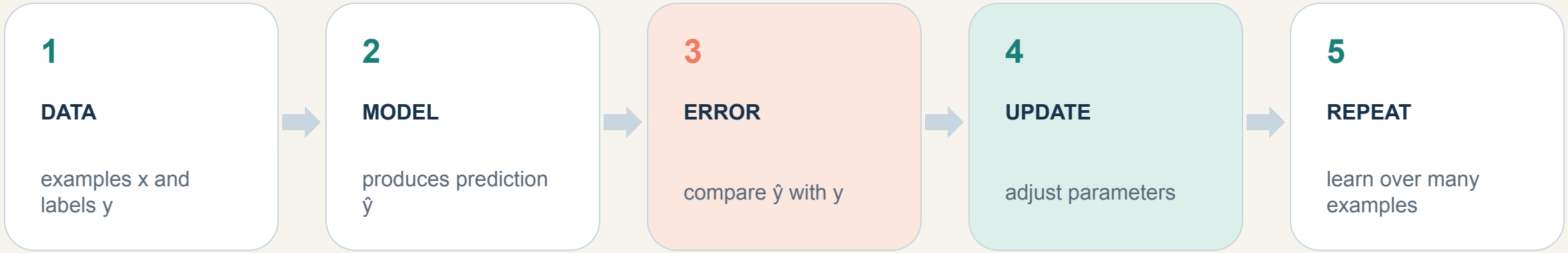
Artificial does not mean “fake”; it means engineered by humans.

Deep learning is one path inside a larger AI landscape



All deep learning is ML, but not all ML is deep learning - and not all AI is ML.

Training changes model parameters to reduce error



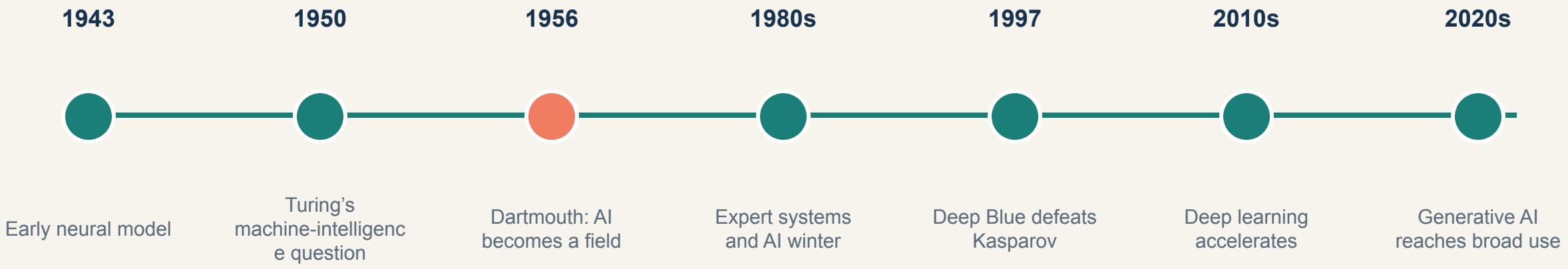
Inference uses the learned parameters; training is when they change.

PART 3

A short history - and what “first” really means

A clean mental model before the details

AI developed through overlapping ideas, tools, and milestones



What counts as the “first AI algorithm”?

1950 - Turing

Proposed a practical test and framed the machine-intelligence question.

1951 - Strachey

Created an early working checkers program.

1955-56 - Logic Theorist

A landmark symbolic theorem-proving program, often taught as an early AI program.

1956 - Dartmouth

The term and research field gained a shared identity.

Teaching answer: Logic Theorist is a landmark; Strachey’s checkers program is even earlier.

Early machine learning also developed in parallel

1949 - Hebb

A conceptual learning principle: connections strengthen when units activate together.

1950s - Samuel

A checkers program improved through experience; closely associated with popularizing “machine learning.”

1958 - Perceptron

An early trainable neural model that learned weights from examples.

Simple class answer: Samuel = early ML system; perceptron = early formal learning algorithm.

Deep learning uses multiple learned layers

Shallow neural network

A smaller number of learned stages

Example: a perceptron or small multilayer perceptron

Can be effective for simpler patterns

Deep neural network

Multiple layers can learn hierarchical representations

Examples: CNNs and transformers

More layers are not automatically better

Data, architecture, optimization, and evaluation still matter.

PART 4

What machines do well - and where they remain limited

A clean mental model before the details

Machine strength is not the same as general intelligence

Machine strengths

Very fast arithmetic and search

Large digital memory and retrieval

Repeat tasks without fatigue

Process enormous datasets

Superhuman performance in some narrow domains

Human strengths

Flexible transfer across very different tasks

Common sense grounded in everyday experience

Adaptation when rules suddenly change

Rich social and contextual understanding

Learning new concepts from little experience

Superhuman at chess does not imply human-level general intelligence.

Current AI does not reliably match broad, robust human intelligence

Common sense

Handle totally novel everyday situations reliably

Continual learning

Learn new things without damaging old knowledge

Embodied understanding

Ground concepts in physical, lived experience

Long-horizon autonomy

Plan, act, recover, and remain aligned over long tasks

Knowing when it is wrong

Calibrate confidence and avoid confident mistakes

Broad transfer

Apply one learned idea across very different domains with little data

Precise wording matters: “does not reliably match humans yet,” not “can never do this.”

Use six questions to decode almost any AI/ML paper

1 Problem

What is the paper trying to solve?

2 Contribution

What is actually new?

3 Data

What inputs, labels, or environment are used?

4 Learning

Rules/search, supervised, unsupervised, RL, or DL?

5 Evidence

Which baselines and metrics support the claim?

6 Limits

Can we reproduce it, and where does it fail?

Research begins when you ask: “What experiment should I run next?”

Optional videos from the original class deck

AI / ML / Deep Learning overview	https://www.youtube.com/watch?v=4RixMPF4xis
History of AI	https://www.youtube.com/watch?v=UFJQV5Jb0pY
Supervised learning	https://www.youtube.com/watch?v=4qVRBYAdLAo
Unsupervised learning	https://www.youtube.com/watch?v=JnnaDNNb380
Reinforcement learning	https://www.youtube.com/watch?v=nIgIv4IfJ6s

THE CORE IDEA

Choose the problem before the model

AI includes many ways to decide. Machine learning is the branch that learns patterns from data.

Frame the inputs, outputs, feedback, and measure of success - then choose the method.